



Lab 04: Router Configuration & Subnetting

Switch0 (2960-24TT) · Router0 (2911) · PC0 · PC1 · PC2

Lab #04

Topic: Subnetting & Routing

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Connect a 2960-24TT switch to a 2911 router via Fa0/3 to Gig0/0.
- Configure the router Gigabit interface as the default gateway for the LAN.
- Assign static IPs to PC0, PC1 and PC2 on the same subnet.
- Connect PC2 directly to the router on Gig0/1 as a separate subnet.
- Verify full connectivity using ping and show commands.

2. Network Topology

Switch0 (2960-24TT) connects PC0 (Fa0/1) and PC1 (Fa0/2) and uplinks to Router0 on Fa0/3 → Gig0/0. PC2 connects directly to Router0 on Gig0/1. Two subnets exist: LAN subnet and a direct PC2 subnet.

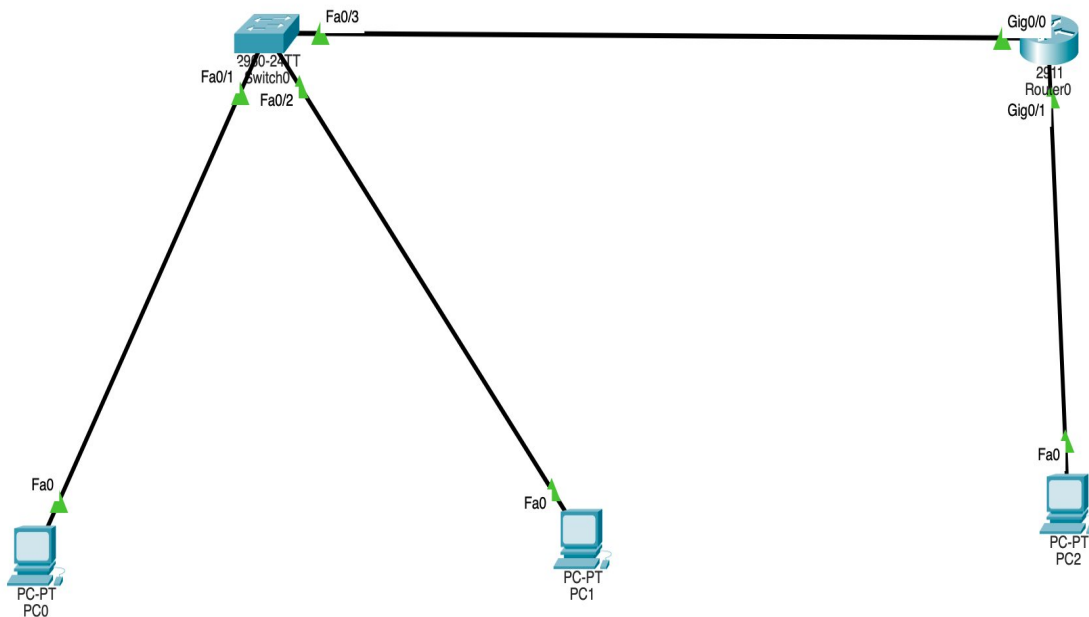


Figure 1 – Topology: Switch0 with PC0/PC1, Router0 (2911), PC2 on Gig0/1

Device	Interface	IP Address	Subnet Mask	Gateway
Router0	Gig0/0 (to Switch)	192.168.1.1	255.255.255.0	—
Router0	Gig0/1 (to PC2)	192.168.2.1	255.255.255.0	—



PC0	Fa0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	Fa0	192.168.1.20	255.255.255.0	192.168.1.1
PC2	Fa0	192.168.2.10	255.255.255.0	192.168.2.1

3. Step-by-Step Configuration

3.1 Router0 – Interface Configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname Router0
Router0(config)# interface GigabitEthernet0/0
Router0(config-if)# ip address 192.168.1.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface GigabitEthernet0/1
Router0(config-if)# ip address 192.168.2.1 255.255.255.0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0# write memory
```

4. Verification

```
PC0> ping 192.168.1.20    ! PC0 to PC1 (same subnet, via switch)
PC0> ping 192.168.2.10    ! PC0 to PC2 (cross-subnet, via router)
Router0# show ip interface brief
Router0# show ip route
```

5. Key Takeaways

- The 2911 router uses GigabitEthernet interfaces — faster and more modern than FastEthernet.
- PC0 and PC1 are on the same /24 subnet so switch-level forwarding handles them without the router.
- PC2 is on a separate /24 requiring the router to forward packets between subnets.
- Connected routes appear automatically in 'show ip route' when interfaces are up.
- The switch (2960) needs no IP for basic Layer 2 forwarding — only for remote management (SVI).